



Normal aging

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Literature review current through: **Aug 2026**.

This topic last updated: **Sep 09, 2025**.

INTRODUCTION

Aging is characterized by progressive and broadly predictable changes associated with increased susceptibility to many diseases. Aging is not a homogenous process. Rather, organs in the same person age at different rates influenced by multiple factors, including genetic makeup, lifestyle choices, and environmental exposures. As an example, mutations accumulate in stem cells with aging but are different depending on the organ [1]. Epigenetic deoxyribonucleic acid (DNA) modifications with age are also highly tissue specific [2]. A Danish twin study found that genetics accounted for approximately 25 percent of the variation in longevity among twins and environmental factors accounted for approximately 50 percent [3]. However, with greater longevity (to age 90 or 100), genetic influences became more important.

This topic will present an overview of normal aging. Effects of aging on the immune systems and abnormal aging are discussed in more detail separately. (See "[Immune function in older adults](#)".)

AGE-ASSOCIATED PHYSIOLOGIC CHANGES

Physiologic rhythms — The organization of rhythmic physiologic processes is altered by aging. Age impacts the circadian pattern of body temperature, plasma cortisol, and sleep and can cause desynchronization or "internal phase drift." Phase advances can lead to the occurrence of some rhythmic functions (eg, the 24-hour body temperature trough and sleep onset) one to two hours earlier in older adults as well as decreases in the magnitude of the

oscillations. In addition, age may delay the ability to reset physiologic rhythms to a new photoperiod.

The pulsatile secretion of gonadotropins, growth hormone, thyrotropin, melatonin, and adrenocorticotrophic hormone (ACTH) are attenuated with age [4]. The diurnal rhythmicity of cortisol is preserved in older age but with a decreased amplitude and delayed elevation [5]. One source of this dysfunction appears to be neuronal loss in the suprachiasmatic nucleus in the hypothalamus [6].

Loss of complexity — Loss of complexity, a concept derived from the field of nonlinear dynamics, may be a general principle of all aging systems [7]. This loss of complexity may result in decreased heart rate variability, blood pressure variability, electroencephalographic frequencies, response to auditory frequencies, and response to stress. Age-related loss of complexity may not be immutable, however; as an example, senior athletes show greater heart rate variability than sedentary age-matched controls [8].

Homeostenosis — Homeostenosis refers to the concept that, from maturity to senescence, diminishing physiologic reserves are available to meet challenges to homeostasis. This concept was first recognized by Walter Cannon in the 1940s [9]. Homeostenosis leads to the increased vulnerability to disease that occurs with aging.

A figure graphically displays the traditional thinking about homeostenosis ([figure 1](#)). The endpoint of this process is frailty, where even the smallest challenge overwhelms the available reserves and results in disaster. The "precipice" may be variably defined: death, cardiac arrest, hospital admission, or onset of a symptom such as confusion or incontinence. Aging itself brings the individual closer to the precipice by the loss of physiologic reserves. With aging, the area in which the older person can bring themselves back to homeostasis by invoking their reserves narrows or becomes stenotic.

Evidence for this model is plentiful. As an example, the Acute Physiologic and Chronic Health Evaluation (APACHE) severity of illness scales, used to predict prognosis for patients in intensive care, have a correction for age. The Acute Physiologic Assessment, a component of the APACHE score, indicates deviation from homeostatic values for 12 variables, including vital signs, oxygenation, pH, electrolytes, hematocrit, white blood count, and creatinine. A zero score indicates homeostasis, and a greater point total indicates a larger deviation from homeostasis [10,11]. In a comparison of young and old patients who had a cardiac arrest, the younger group (mean age 59) had significantly higher Acute Physiologic Assessment scores in the 24 hours pre-arrest than the older group (mean age 75) [12]. These data indicate that the deviation from homeostasis needed to cross a critical threshold (cardiac arrest) is less in the old. In practical terms, the creators of the APACHE scales recognize this by giving "age points" so that total scores are equalized between the groups. (See "[Predictive scoring systems in the intensive care unit](#)", section on 'Acute Physiologic and Chronic Health Evaluation (APACHE)').

Maintaining homeostasis is a dynamic, active process. Frailty is the state when physiologic reserves are maximally invoked just to maintain homeostasis and the most modest challenge impels them across some threshold. Physiologic and functional measures are more likely to be out of "normal range" with normal aging, and the fraction of measures that are out of normal constitutes the Frailty Index [13]. The Frailty Index increases with age. Increased severity of illness and frailty have independent effects on patient outcomes [14]. Physical resilience is defined as the ability to withstand or recover from the functional impact caused by acute health stressors. It is not the opposite of frailty nor homeostenosis but, because late-life stressors are inevitable, may define successful aging [15]. (See "Frailty".)

The "family of precipices" concept is useful in understanding altered presentations of disease in older adults ([figure 1](#)). As an example, delirium is a common presentation of a wide variety of illness in the older individual, a marker of the uneasy truce that the old brain maintains with the environment. A given older person may have the same presentation (confusion) for a urinary tract infection, gastrointestinal bleeding, or a myocardial infarction. The systemic responses to these differing illnesses may be similar, involving catecholamines and mediators of inflammation. The "anti-confusion reserves" are exhausted, so the distance from homeostasis to this "precipice" is easily crossed.

In summary, an apparent loss of physiologic reserves in older adults leads to intolerance to challenges to their homeostasis. This increased vulnerability is in part because the older person is continually expending reserves to compensate for primary age changes as well as other processes that are absent or trivial in the younger individual. Thus, older adults have increased frailty and vulnerability as well as decreased robustness and resilience.

Nonlinearity of aging — Aging is not linear and occurs at a faster rate at different ages. An analysis of plasma markers reveals distinct patterns of dysregulated molecules, indicating two peak ages where major dysregulation in aging occurs: at 44 years old, and again at 60 years old [16].

HEMATOPOIETIC SYSTEM

In the absence of additional challenges, the hematopoietic system maintains adequate function throughout an individual's lifespan [17]. Red cell lifespan, iron turnover, and blood volume are unchanged with age. However, bone marrow mass decreases and fat in the bone marrow increases with age [18,19]. This increase in fat is not uniform; it is more extensive in the femoral head than the shaft [20]. Hematopoietic functional reserves are reduced with age. Thus, advanced age may be an important consideration for determining suitability as a donor for hematopoietic cell transplantation [21] or tolerance of chemotherapy.

The compensatory hematopoietic response to phlebotomy, hypoxia, and other challenges is delayed and less vigorous in the older person [22]. This is due to changes to populations of progenitor cells, and to the bone marrow environmental matrix [23]. As an example, studies comparing the bone marrow of healthy older and younger patients found a 35 percent decrease in the colony size of the stimulated erythroid progenitor (colony-forming unit-erythroid [CFU-E]) cells in older individuals [24]. Another study suggests that inability to produce critical stimulatory hormones (stem cell factor, granulocyte macrophage colony-stimulating factor [GM-CSF], and interleukin 3 [IL-3]) is the major factor accounting for functional difference between bone marrow from older and younger people [25].

Analysis of hematopoietic precursor cells from older adults reveals a dramatic decrease in genetic diversity, possibly related to genetic drift or environmental changes [26]. The frequency of clonal hematopoiesis of indeterminate potential (CHIP) and its associated mutations reaches 50 percent in those over 90 [27]. In addition to having an increased risk (0.5 percent per year) of development of hematologic malignancies [28], age-related CHIP is associated with increased mortality from many causes [26]. One relatively common finding in older men is loss of the Y chromosome from a proportion of leukocytes as a clone marker [29]. The presence of this clone, as with many of the other mutations defining age-related CHIP, is associated with increased mortality from both cancers and nononcologic processes [30,31].

While physiologic aging does not result in anemia, the risk of anemia is increased by a number of mechanisms with age. Hepcidin, an iron regulatory hormone, is stimulated by inflammatory cytokines. Hepcidin blocks iron release from macrophages and hepatocytes (which recycle it from hemophagocytosis) and inhibits iron transport from enterocytes in the gut to the plasma. Thus, inflammaging, the persistent low-level proinflammatory state in normal older adults, impairs their response to erythropoietin or similar stimuli [32].

Total circulating white cell counts do not change with age in healthy older people, but the function of several cell types is reduced. Age-related changes in the immune system are discussed separately. (See '[Immune system](#)' below and '[Immune function in older adults](#)'.)

Age is a significant risk factor for myelotoxicity due to chemotherapy regimens for malignancies [33]. However, consistent with many other changes, heightened sensitivity to chemotherapy and impaired recovery of the bone marrow is not a uniform finding in all older adults. (See '[Systemic chemotherapy for cancer in older adults](#)'.)

Though the number of platelets is unchanged with age, platelet responsiveness to a number of thrombotic stimulators is increased. Reduced availability of nitric oxide and increased oxidative damage have been implicated in age-associated platelet hyperresponsiveness [34]. This results in a small but consistent decrease in bleeding time with age. Fibrinogen, factor V, factor VII, factor VIII, factor IX, high molecular-weight kininogen, and prekallikrein increase

with age in healthy humans, possibly related to the low-grade inflammation that is part of normal aging [35]. Fibrin degradation fragments (D-dimers) are elevated twofold in healthy older subjects with no evidence of thrombosis and may be even higher in hospitalized older adults, such that an age-incorporating formula for D-dimer interpretation has been suggested [36]. Plasminogen activator inhibitor 1, the major inhibitor of fibrinolysis, increases dramatically with aging [37-39]. Thus, old age should be considered a procoagulant state, and age is an important risk factor for deep venous thrombosis.

GASTROINTESTINAL TRACT

The overall effects of aging on the gastrointestinal system are modest; aging itself does not cause malnourishment. Nonetheless, age-related changes in gastrointestinal systems affect the incidence and presentation of multiple gastrointestinal problems in older adults.

Oropharynx — The epithelial lining of the oral mucosa thins with age. The gums recede, exposing the tooth cementum, which is more prone to decay, and predisposing older persons to root caries and incomplete mastication [40]. Edentate patients are at greater risk for inadequate nutritional intake compared with those with partial or full retention of their teeth [41].

Modest age-associated changes occur in the salivary glands, including a small decrease in the number of acinar cells, and up to a 50 percent decrease in maximal saliva production from parotid salivary glands [42]. Although accessory salivary gland production is unchanged, fatty infiltration of these glands increases with age [43], making the discrimination between Sjögren's disease and age-associated dry mouth more dependent on the extent of fibrosis than fat [44]. Up to 50 percent of older patients have subjective complaints of dry mouth, which can impact chewing and swallowing. Age-related changes in the composition of saliva will modify the adhesion and viscosity of the oral salivary film [45]. However, some of these complaints may be attributed to medication side effects rather than aging itself [46].

Transfer of the food bolus to the pharynx is altered in the majority of older patients. Loss of esophageal muscle compliance results in increased resistance to flow across the upper esophageal sphincter [47]. Up to 60 percent of older patients without dysphagia have abnormal transfer to the pharynx on videofluoroscopy [48]. The strength and coordination of the tongue is impaired in healthy 80-year-old individuals [49]. Less effective mastication and decreased food clearance from the pharynx lead to increased aspiration risk in older adults. Additionally, protective aerodigestive reflexes in the pharynx stimulated by injecting water into the mouth are frequently absent in older people [50].

Structural and functional changes in the oropharynx may impact the mechanisms of obstruction in patients with sleep apnea. Older patients had high incidence of collapse in the

velum region and much lower oropharyngeal lateral wall collapse compared with younger patients [51].

Esophagus — Anatomic changes in the esophagus include hypertrophy of the skeletal muscle at the upper third, decrease in myenteric ganglion cells that coordinate peristalsis, and perhaps increased smooth muscle thickness [40]. The amplitude of esophageal contractions during peristalsis decreases, but the movement of food is not impaired. However, using advanced manometric techniques, more subtle abnormalities in liquid propagation into the stomach can be seen in 40 percent of older adults [52]. Upper esophageal sphincter and esophageal body pressure responses to large-volume simulated reflux are preserved in healthy 70-year-olds, but responses to lower-volume, slowly applied reflux fluid and associated postreflux residue are reduced in elderly individuals, with significant implications for aspiration risk [53]. Abnormal peristalsis after swallowing and non-peristaltic repetitive contractions, at one time attributed to old age and called "presbyesophagus," are now thought to be due to disease processes.

Secondary contractions contribute to clearance of refluxed food or acid. Diminution of these contractions, combined with decreased lower esophageal sphincter tone, results in increased gastric acid exposure [40]. Secondary esophageal contractions induced by esophageal distention and acid infusion into the esophagus appear to be greatly reduced with age [40,53]. Sensation of distention, and possibly tissue damage, in the distal esophagus is also impaired with age [54]. Thus, many older patients with severe reflux esophagitis seen at endoscopy have surprisingly little symptomatology.

Stomach — Early studies suggested that gastric acid production decreased dramatically with age, with a decrease in parietal cells and an increase in interstitial leukocytes [55]. Subsequent studies challenge those findings and suggest that 90 percent of people aged 65 and over are able to acidify gastric contents in the basal unstimulated state [56,57]. *Helicobacter pylori* infection may account for the discrepancies between early and more recent work [58]. Over 50 percent of older people are infected with *H. pylori*, with the prevalence increasing with advancing age [59]. (See "[Helicobacter pylori: Epidemiology, pathophysiology, and overview of disease associations](#)", section on 'Prevalence'.)

Increased rates of gastritis and increased sensitivity to gastric irritants, such as nonsteroidal antiinflammatory medications or bisphosphonates, in older adults may be related to several age-related physiologic changes: decreased prostaglandin synthesis, decreased bicarbonate and nonparietal fluid secretion, delayed gastric emptying, and impaired microcirculation [60]. In addition to an increased sensitivity to gastric insults, rates of healing are impaired by a host of mechanisms in older people [61]. Gastric motility is determined by the combined effects of the enteric nerves, smooth muscle, and the interstitial cells of Cajal. The number and volume of the interstitial cells of Cajal bodies decreases by over 10 percent per decade in normal

people without motility complaints [62]. In aging rat models, sensory neural function is decreased, delaying recognition of experimentally induced mucosal injury [40].

The stomach also has critical endocrine functions. Serum levels of ghrelin and gastrin, as well as ghrelin signaling, are reduced in healthy older adults [63,64].

Small intestine — The small intestine undergoes modest anatomic changes, including moderate villus atrophy and coarsening of the mucosae. The absorption of several micronutrients (eg, xylose, folic acid, B12, copper) may decrease with age but remain adequate for homeostasis [65]. The efficiency of calcium absorption from the gut lumen decreases because of decreased vitamin D receptors in the gut and decreased levels of circulating 25(OH) vitamin D. Typically, women over age 75 absorb 25 percent less of a given dose of calcium than younger women, especially if there is reduced acid secretion [66]. Iron may also be less well-absorbed, but overall aging impacts the absorption of macronutrients minimally [65].

Consumed carbohydrates result in significantly more hydrogen excretion in the older adults, suggesting malabsorption and subsequent bacterial metabolism of the carbohydrate in the aging gut [67]. Up to 15 percent of residents in senior congregate housing have evidence of bacterial overgrowth as assessed by breath hydrogen testing [68]. Bacterial overgrowth and associated malabsorption can affect nutritional status and micronutrient absorption. Additionally, the barrier function of the small intestine may be compromised and local inflammation activated in response [69].

Decreases in sensory and myenteric neurons contribute to the increased frequency of painless ulcers with increased age [40,70]. Interestingly, in several animal studies, caloric restriction has been shown to decrease myenteric neuronal loss with age [71]. Small intestinal transit time (measured with capsule endoscopy) seems to be unchanged by age [72].

Older people require lower radiation doses to develop acute radiation syndrome or to die from radiation, due to increased sensitivity of the small bowel to radiation [73].

Large intestine — Anatomic changes with aging in the large intestine include mucosal atrophy, cellular and structural abnormalities in the mucosal glands, hypertrophy of the muscularis mucosa, and atrophy of the muscularis externa. Functional changes include altered coordination of contraction and increased opioid sensitivity that may predispose the older person to drug-induced constipation.

Studies have not been consistent regarding alterations in colonic motility, but the general consensus is that colonic propulsive motility is reduced with age, with colonic transit time measured by capsule endoscopy being almost doubled in community-dwelling older adults compared with young people [74]. Approximately one-fourth of those over 65 years old suffer from chronic constipation [41]. One factor contributing to reduced motility is an age-related

reduction in myenteric plexus neurons and a decline in the interstitial cells of Cajal similar to that seen in the stomach. Intrinsic sensory neurons that respond to physicochemical changes may degenerate disproportionately compared with motor enteric neurons. The loss of sensory input into local reflex pathways could contribute to reduced propulsive motility [75]. (See ["Etiology and evaluation of chronic constipation in adults"](#).)

The loss of intrinsic sensory neurons may also contribute to the decreased visceral response, including decrease in perceived pain with bowel perforation, distention, or ischemia [76]. As an example, the rigid surgical abdomen after appendiceal perforation is a less frequent finding in those over 75, leading to delayed diagnosis [77].

Older women may be more predisposed to fecal incontinence than older men as the resting pressure and squeeze pressure decrease with age, resulting in decreased anal sphincter tone [78]. In one study, both male and female patients aged greater than 70 years had 30 to 40 percent decreases in sphincter pressures compared with controls less than 30 years of age [79]. The internal anal sphincter of continent older people is thickened, perhaps to compensate for decreased resting and maximum pressures in the anal canal with age. However, thinning of the external sphincter correlated with fecal incontinence more than age [80].

Diverticula are common in Western populations over age 65, with prevalence ≥ 65 percent [81]. The prevalence of diverticula is lower in other populations, presumably with other diets, but nonetheless there remains a strong age-dependence [82]. The formation of colonic diverticula is attributed to decreased muscle wall strength, decreased bowel wall compliance, and increased intraabdominal pressure required for stool excretion [81]. Slower large bowel transit and increased segmental contractions (as opposed to propulsive contractions) result in increased water reabsorption, leaving harder stools and increasing the likelihood of wall failure [40]. (See ["Colonic diverticulosis and diverticular disease: Epidemiology, risk factors, and pathogenesis"](#).)

The risk of colon cancer increases with age. In addition to prolonged exposure to potential carcinogens, aging is associated with increased proliferation and decreased apoptosis in the colonic mucosa [83]. The biology of these changes is a rich area for exploration. (See ["Epidemiology and risk factors for colorectal cancer"](#).)

The barrier function of colonic epithelium may be compromised and has been implicated in promoting the proinflammatory state, "inflammaging." Isolated specimens from normal aging baboon colon showed increased permeability to potential toxins, decreases in the key structural components of the tight junctions, and enhanced downstream inflammatory responses [84]. The deterioration in the barrier function of the intestine with age appears to occur across species and has been implicated in primary aging changes like thymic evolution [85,86]. By contrast, barrier function for [lactulose](#) and sucrose is preserved in healthy 65- to

75-year-old persons [87] but, for many other chemicals, the effects of age in healthy older adults are unknown.

The gut microbiome changes with healthy aging and with age-associated diseases. Some have proposed a "healthy signature" based on stool analysis [88], while others suggest that it is the gut biome through its effect of metabolites and inflammation that promotes healthy aging [89]. The immune system of the intestinal mucosa, a barrier for many products of the microbiome, is compromised with aging [90]. Thus, the extent with which an older adult interacts with their microbiome may be increased.

Hepatobiliary system — Liver mass decreases between 20 and 40 percent with age, and liver perfusion and blood flow decreases up to 50 percent between the 3rd and 10th decades of life [91]. Lipofuscin accumulates in hepatocytes with age and is also seen in young patients with severe malnutrition, accounting for an appearance that has been described as "brown atrophy." Older livers have more macrohepatocytes (large cells) and increased polyploidy [92]. The number of mitochondria per cell increases with age [93]. The older liver is less tolerant of ischemia which increases the risk of using older livers for transplantation [94]. However, with adequate screening criteria, livers from those over 80 can have excellent outcomes after transplant [95].

The following findings are relevant to liver function in older adults:

- Although many liver functions decline (diminished erythromycin demethylation, galactose elimination, and reduced caffeine clearance), standard "liver function tests" (transaminases, alkaline phosphatase) are minimally affected by age [40,96].
- Findings are contradictory regarding albumin synthesis in older livers; animal studies found reductions consistent with a loss of liver mass [97], although this was not confirmed in a study in healthy older people [98]. Serum albumin declines slightly with normal human aging [99]. Interestingly, studies have shown that mortality of nursing home residents correlates with albumin levels, even within the normal range [100].
- The metabolism of low-density lipoprotein (LDL) cholesterol decreases with a reduction in LDL receptors in older patients [101], which could contribute to the higher serum LDL levels in older adults [97].
- Cytochrome P450 content decreases with age, with one study finding a 32 percent decrement comparing individuals over 70 years with a group 20 to 29 years of age [102]. This may account for the finding that metabolic clearance of many drugs is 20 to 40 percent slower in older people [103]. (See "Drug prescribing for older adults".)
- The lower amounts of vitamin K antagonists needed to anticoagulate older people are consistent with age-related decreased synthesis of vitamin K-dependent clotting factors

[104].

- Although function and anatomy of the gall bladder are well preserved in old age, the bile composition has a higher lithogenic index, predisposing the older person to cholesterol gallstone formation [105].

Younger livers show a robust regenerative response to liver injury characterized by mitogen-activated protein kinase activity, which declines with age [101]. One consequence of this impaired liver regeneration is that a larger remaining liver is required for older people after major hepatectomy [106].

Exocrine pancreas — The exocrine pancreas undergoes only modest alterations with age. Minor atrophic and fibrotic changes have essentially no impact on pancreatic exocrine function [67]. The fat fraction of the pancreas increases with age in healthy women [107]. Noninvasive pancreatography indicates main pancreatic duct dilatation, greater incidence of cysts and side branches of the pancreatic ducts [108], and a decrease in stimulated pancreatic flow with advancing age [109]. Aged animals showed decreased output of lipase and amylase in response to meals that were high in fat or carbohydrates [40].

RENAL SYSTEM

There are multiple effects of aging on the renal system ([table 1](#)). Kidney mass decreases by 25 to 30 percent between the ages of 30 and 80 years, with the steepest decline after age 50. In addition, fat and fibrosis replace some of the remaining functional parenchyma. Loss occurs primarily in the cortex and preferentially affects those nephrons most important to maximal urine concentration. Senescent cells are more common with increasing age in donor kidney cortices [110], suggesting that senolytic agents may have a potential role in improving age-related declines in renal function [111]. As the senescent cells produce proinflammatory as well as profibrotic signals, the extent of fibrosis, extracellular matrix, and inflammation increases with age [112].

Normal aging is associated with a reduction in functional glomeruli of almost 50 percent, as found in a comparison of kidneys from donors aged 18 to 29 with those aged 70 to 75 years. The number of sclerotic glomeruli increased almost 10-fold across the same groups [113]. Atrophy and resorption of nephrons may contribute to the age effect more than diffuse sclerosis of glomeruli [114]. The remaining glomeruli have impaired filtering ability, though single nephron studies showed preserved filtration rate until age 70 [115]. Intrarenal vascular changes include spiraling of the afferent arterioles, narrowing of the larger arteries, intimal fibrosis [116], and shunts between afferent and efferent arterioles allowing blood flow to bypass the glomeruli [117]. Nephrosclerosis (global glomerulosclerosis, interstitial fibrosis, and arteriosclerosis) was identified in donor kidneys to be used for transplantation in 3

percent of donors 18 to 29 years old and in 73 percent of donors 70 to 77 years old [118]. Evidence of age-associated renal remodeling is confirmed with proteomic studies of urine from healthy individuals [119].

At baseline, kidney plasma blood flow is 40 percent lower in healthy normotensive older men than in young men, and this difference is magnified under conditions that stimulate renal vasodilation [120]. Studies suggest that older kidneys may be maintained in a state of vasodilation to compensate for loss of vasculature [120,121]. Vasodilating prostaglandins are increased at baseline in normal older adults [122], and this contributes to the increased (roughly doubled) risk of kidney injury with use of nonsteroidal antiinflammatory drugs (NSAIDs) in older people [123,124].

Creatinine clearance decreases with age (7.5 to 10 mL per minute per decade), although there is wide variability in decline seen in longitudinal studies of healthy older adults [125]. As many as one-third of older adults monitored for over a decade have no change at all (half those actually improve) in the glomerular filtration rate (GFR), one-third have a slight decline, and one-third have a more marked decline. Creatinine production also decreases with age and tubular secretion of creatinine increases, so that the serum creatinine may remain stable despite decreases in the GFR [126]. Hypertrophied and compensating nephrons are often less tolerant of additional insults thus more prone to acute kidney injury [127]. All of the commonly used equations for estimating creatinine clearance factor age into the formulae; however, the estimated GFR (eGFR) provided by some electronic health records need to be utilized cautiously, especially with those over 90 [128,129].

Cystatin C-based estimates of kidney function may be useful when accurate assessment in an older person is necessary [130]. In healthy older people, an increase of roughly 50 percent in cystatin C levels is seen from age 40 to age 80 [131] (see "[Calculation of the creatinine clearance](#)"). Increases in cystatin C track robustly with functional decline in longitudinal studies [132]. More physically active older men had greater preservation of GFR [133].

Fluid and electrolyte homeostasis are maintained relatively well with aging, in the absence of challenges. However, the ability to maximally dilute urine and excrete a water load is impaired and compromises volume regulation under conditions of stress. In the setting of dehydration, the minimum urine flow rate is twice as great in those over 70 compared with those under 40, and the maximum urine osmolality is also reduced with age [134]. In addition to this impaired ability to retain water and solute, the older kidney also is impaired in its ability to retain amino acids and glucose.

Other functional changes in the renal system are a reduction of urine acidification and impairment in excreting an acid load. The older kidney is more prone to nephrotoxicity related to medications, chemotherapy, or intravenous contrast [135,136]. Additionally, the injured older kidney is less likely to recover from acute insult [137]. The older kidney is also more

vulnerable to ischemic insult, with a greater number of cells undergoing apoptosis following ischemia than in the young kidney. Tubular cells appear to have diminished ability to entirely repopulate the tubules after an acute ischemic insult, potentially an effect of the increasing senescent cell population of the aging kidney [111].

Hormonal functions of the kidney are affected by aging, including decreased hydroxylation of vitamin D [138,139]. In older women, the vitamin D response to parathyroid hormone infusion is attenuated [140]. Downregulation of the renin-angiotensin aldosterone system is seen in normal and hypertensive older people [141-143]. However, the autonomous release of aldosterone by aldosterone-producing cell clusters is increasingly common with age, at least to age 70 [144]. This leads to lack of suppression of aldosterone, with salt loading and inadequate increases in response to sodium restriction [144]. The production of erythropoietin in response to hemoglobin, however, appears to be unchanged with age [145]. However, others have reported the circulating erythropoietin levels in older adults with anemia is lower than young adults with the same level of anemia [146]. Finally, the kidney is a primary source for klotho, a membrane protein involved in phosphate metabolism that may have important implications for aging [147]. The kidneys' production of klotho decreases with increasing age and genetic polymorphisms that increase its level are associated with increased longevity [148].

CARDIOVASCULAR SYSTEM

Advancing age increases the risk for hypertension and coronary artery disease. The prevalence of coronary artery disease at autopsy may reach 75 percent after the sixth decade in men and two decades later in women [149]. Therefore, to isolate age-related cardiovascular change from disease-related change, studies must carefully select older individuals with no underlying cardiovascular condition. The Baltimore Longitudinal study studied highly screened older individuals and found only a minimal impact of aging on resting cardiovascular function such as left ventricular ejection fraction (LVEF) [150]. This reflects the adequacy of the compensatory strategies used by the old heart (and vascular system) to counteract subtle and gradual age-associated physiologic, molecular, and biochemical changes. However, by invoking available compensatory mechanisms to maintain resting function, the older person is less able to compensate for subsequent challenges [151]. (See 'Homeostenosis' above.)

Many older people perform little physical activity. Typical age changes may therefore also reflect the impact of factors related to lifestyle and comorbidity, and the contribution of age alone may be difficult to determine. Physical exercise may mitigate some of the age-related changes.

Modest anatomic changes occur in the right side of the heart. Right atrial volume increases modestly; however, mean and peak systolic blood flow in the superior and inferior vena cavae decrease with age [152]. By contrast, the left atrium enlarges, and the left ventricle stiffens with aging. Left atrial volume, corrected for body size, increases roughly 50 percent from the third decade to the eighth [153]. The larger left atrium becomes stiffer, less able to perform as a reservoir [154,155]. Age also increases the risk of left atrial cardiomyopathy significantly [156]. The left ventricle also hypertrophies with age, with an average increase in left ventricular wall thickness of 10 percent [157]. In healthy women, the left ventricle end diastolic volume decreases by 10 to 15 percent from age 20 to age 80 [158]. A group of highly screened, healthy older people had more modest changes with age. With exercise, at the same filling pressures, older adults had smaller left ventricular volumes, suggesting the diastolic stiffness is also important in limiting maximum cardiac output [159].

Both the aortic valve and mitral annulus thicken and develop calcific deposits [160]. Mitral annular calcification may predispose the older person to cardiac conduction problems. (See "[Valvular heart disease in older adults](#)".)

Ventricular cardiomyocytes hypertrophy, in part as a response to the increased afterload produced by large artery stiffening [161,162]. The largest myocytes are also the most vulnerable to challenge [162]. Loss of myocytes with age has been reported to occur by both apoptosis and necrosis; the total number of cardiomyocytes may be reduced significantly in healthy human hearts [161,163,164]. The loss of myocytes is compensated for by cell hypertrophy, with no net loss in cardiac mass. As well, substantial cellular dropout occurs in the sinoatrial (SA) node and more modest cellular loss at the atrioventricular node. This may underlie increased sensitivity of the older SA node to calcium channel blockers [165].

There is a negligible age-related decrease in the resting heart rate but a marked decrease in the maximum heart rate in response to exertion or other stressors. The intrinsic heart rate (the rate without sympathetic or parasympathetic input to the heart) decreases by five to six beats per minute each decade. The response to both parasympathetic antagonists ([atropine](#)) and beta-adrenergic agonists ([isoproterenol](#)) is decreased in healthy older people [166]. The energy state (phosphocreatine to ATP ratio) of the older heart is diminished, and correlates with poorer diastolic function and lower peak work [167]. Cannabinoid receptors, both CB1 and CB2, are reduced in hearts from people older than 50 [168], which may attenuate some of the cardiovascular effects of cannabinoids in older people [168,169].

Heart rate reflects the combined effects of sympathetic and parasympathetic tones. The target maximum heart rate is calculated as "220 – age." Women may have a more gradual decline and a correction factor of 0.85 to 0.90 may adjust the target heart rate for women. Exercise training does not modify the age-associated decline in maximum heart rate [170]. Heart rate variability, perhaps due to decreased parasympathetic tone and decreased sympathetic responsiveness, also decreases with age [171].

The prevalence of premature atrial complex (also referred to a premature atrial beat, premature supraventricular complex, or premature supraventricular beat) increases with age but is not associated with increased cardiac risk [172]. An increase in isolated ventricular ectopic beats is also seen in healthy older individuals [173].

The culmination of age-associated cardiovascular changes is a decrease in maximum work, measured as maximum oxygen utilization (VO₂max) on exercise testing. Exercise training in sedentary older individuals can improve this parameter, but a decline with age in parameters such as maximum heart rate is seen even in highly fit individuals [170].

Resting LVEF is not changed in healthy older people, but there are smaller increases in LVEF in response to exercise [174]. At maximum effort, LVEF in the young is above 80 percent, while by age 80 it is 70 percent [150]. Older hearts also have impaired early left ventricular filling with a compensatory greater contribution from atrial systole than younger hearts [175]. This impairment in diastolic function reflects multiple age-related changes from decreased sarcoplasmic reticulum calcium uptake due to alterations of the pump protein (SERCA 2a) [176], increased sarcoplasmic reticulum leak, reduced energetics [177], and increased cardiac interstitial fibrosis [178,179].

This diastolic dysfunction and increased reliance on atrial systole may in part explain why atrial fibrillation is more likely to precipitate heart failure in older adults. An atrial gallop (S4) is a normal finding on physical examination in individuals in sinus rhythm over 75, a manifestation of the increased contribution of left atrial systole to ventricular filling. Perhaps as a marker of the increased filling pressure at rest, brain natriuretic peptide (BNP) is increased with age in healthy people [180]. With exertion, healthy older people have a greater increase in pulmonary capillary wedge pressure, reflecting a reliance on Starling's law to increase cardiac output [181].

The old heart is a vulnerable heart. For example, mortality and the probability of developing heart failure after a myocardial infarction increase dramatically with age. While myocardial infarction is not a part of normal aging, response to this systemic challenge is impaired because of the aging process. Similarly, doxorubicin-related heart failure occurs with greater frequency and at a lower cumulative dose in those over 65 [182].

Large arteries are altered with age. The aorta increases in diameter, with the upper limits of normal increasing by approximately 5 mm from age 20 to 40 compared with those over 60 [183]. Length increases a few cm from age 20 to 80 [184]. The aorta increases stiffness as measured by pulse wave velocity (PWV) two- to threefold. This means that pressure waves generated when the aortic valve opens are reflected and return to the heart before the aortic valve is closed, thus increasing the load on the heart. In youth, the reflected waves return after the aortic valve is closed and help perfuse the coronary arteries. In serial studies using four-dimensional magnetic resonance imaging (MRI), the increases in aortic PWV were

greater in the older groups (70 to 80) of males and females up to 10 percent over six years [185]. While PWV requires special equipment to measure, a similar pattern is found in pulse pressure (systolic-diastolic) [186]. Critically the increase in arterial stiffness, especially in the carotids, correlates with decreasing cerebral blood flow in cognitively normal older adults [187]. The effects of disease such as hypertension can double the rate of stiffening [188] and chronic exercise can attenuate it. The ability of the brachial artery to dilate in response to a heat load is relatively preserved while the lower extremity in the older person is diminished [189].

RESPIRATORY SYSTEM

Aging, in the absence of additional challenges, does not result in hypoxia or pneumonia. However, age-related anatomic and functional changes in the respiratory system contribute to the increased frequency of pneumonia, increased likelihood of hypoxia, and decreased maximum oxygen uptake in the older person.

The lung undergoes a number of anatomic changes [190]. Alveolar ducts enlarge due to loss of elastic tissue and alterations in the supporting network of collagen fibers, resulting in a decreased surface area for gas exchange. Overall, approximately one-third of the surface area per volume of lung tissue is lost over the life span, and anatomic dead space increases [191]. The loss of lung elastic tissue decreases recoil and results in modest reduction in the expiratory boundary of the maximal flow-volume envelope. The changes in elastic recoil are not uniform and therefore contribute to the heterogeneity in ventilation and the VQ mismatch described below [192]. During maximal exercise, this may limit expiratory airflow and produce dynamic lung hyperinflation [193]. Surfactant composition is also altered by age [194], and alveolar fluid has a greater content of proinflammatory proteins and a reduced antiinflammatory profile [195]. Lung angiotensin-converting enzyme 2 (ACE2) expression (the severe acute respiratory syndrome coronavirus 2 [SARS-CoV-2] virus receptor) increases with age [196]. Many of these changes in lung function are driven by senescent cells [197]. Carbon monoxide diffusion studies find that diffusion capacity decreases approximately 5 percent per decade [198], although this may be less in highly fit older individuals [199].

Age increases ventilation-perfusion mismatching because airways in dependent portions of the older lung, areas that are better perfused than elsewhere, are closed during all or part of the respiratory cycle. This is a critical factor in the declining arterial PO_2 with age. Alveolar PO_2 does not change with age, but age increases the alveolar-arterial (A-a) oxygen gradient. The effect of the ventilation-perfusion mismatch is more marked in the supine than sitting position because of positional changes in thoracic mechanics [200]. The decrease in arterial PO_2 (PaO_2) may not be linear but appears to decline from age 30 until 70 or 75 and thereafter remains almost constant. While age-related changes do not result in hypoxia at sea level, older adults may approach hypoxia at altitude [201]. The fall on PaO_2 is slightly greater in

women than men [200,202]. Older people may have little reserve for further decrements in pulmonary function before important hemoglobin desaturation occurs.

In contrast to the decrease in PaO_2 and increase in the A-a oxygen gradient, carbon dioxide excretion is not impaired with age; changes in PaCO_2 are due to disease and should not be attributed to age alone [203].

The chest wall also changes with age; increased stiffness of the chest wall predominates over an increase in compliance of the lung parenchyma. Overall chest wall compliance decreases by one-third from age 30 to 75 [204]. Intercostal muscle contraction accounts for less chest expansion in older individuals, with a relatively greater contribution from abdominal muscles. Abdominal muscles are only partially effective in ventilating in the seated (or supine) position. Thus, full airway expansion occurs only in the standing position in older adults. Atelectasis can result in an increased A-a gradient.

As the chest remodels with age, the diaphragm flattens and becomes less efficient. The diaphragmatic changes likely contribute to the increase in the work of breathing during exercise, which can increase 30 percent [205]. Using simulations that model mechanical ventilation, the older person has to work harder to breathe, which may contribute to difficulty in weaning from the ventilator [206].

The effect of age on traditional pulmonary function tests is shown in the graph ([figure 2](#)). With advancing age, functional reserves decrease. In nonsmoking men, forced vital capacity (FVC) decreases between 0.15 and 0.3 liters per decade, and forced expiratory volume in one second (FEV_1) decreases by 0.2 to 0.3 liters per decade, with steeper decline in the seventh and eighth decades [207,208]. Age-related changes in women decline less steeply. The decreases in pulmonary function tests correlate with decreases in skeletal muscle strength, as measured by grip strength [209]. The decrease in FEV_1 is accelerated after menopause [210] and may be reduced by long-term hormone replacement therapy [211] and has been reduced by a decrease in second-hand smoke exposure [212]. Total lung capacity (proportional to height) does not change significantly with age; however, the residual volume (air left in the lung at the end of full expiration) increases by as much as 10 percent per decade because of a higher closing volume. Of note, however, the ability to properly perform pulmonary function tests (and use inhalers) decreases in older adult patients [213]. Finally, the correlations between spirometry and diffusion parameters is poor in healthy older people, suggesting that lung aging is not a uniform process [214].

The Cardiovascular Health Study population experienced age-related decreases in maximal inspiratory pressure (MIP) or force, and a smaller decrement in maximal expiratory force parameters [215]. The decline in MIP is linear until age 65 but appears to accelerate afterwards [216]. Both the inspiratory force and expiratory force are significantly greater in physically active older people than sedentary ones. Diaphragm thickness was also greater in

the active old group [217], but some of the apparent decrease in diaphragm strength may be due to the loss of curvature, the result of the chest wall changes described above [218]. Nevertheless, some of the decrements described above are due to sedentary lifestyle.

Older persons have decreased responses to hypoxemia, hypercapnia, and mechanical loading, such as breathing through a small-diameter endotracheal tube [219]. The central drive to the respiratory muscles is decreased [220]. Many of these changes are minimized with exercise; the implication is that central or peripheral receptor hypo-responsiveness may be due in part to deconditioning, and that exercise training can induce compensation for age-related changes. Older adults recover more slowly from insults that damage or destroy lung tissue [221].

Cough is less vigorous in the older person because of the age effects on respiratory muscle strength and greater closing volumes that prevent clearing of increasing proportions of the lungs [215]. Mucociliary clearance is slower and less effective [222], and recovery of mucociliary clearance after insult (typically viral infection) is slowed with age. In addition to impaired clearance from large airways, clearance of inhaled particles from the small conducting airways is also impaired with age [223]. This may lead to changes in the lung microbiome as loss of diversity correlates with lower FEV₁ in middle aged and older people [224]. The cellular changes in the lung epithelium, especially those prompted by chronic inflammation, lead to a thinner epithelial layer and altered receptor expression, both increasing the risk of pneumonia [225].

The rate of decline in the peak aerobic capacity in healthy older persons with age, especially men, is not constant. The Baltimore Longitudinal Study of Aging found a 3 to 6 percent decrease in peak aerobic capacity per decade in the 30s and more than 20 percent decrease in peak aerobic capacity per decade in the 70s and beyond [226]. The decrease in FEV₁ with age correlates strongly with the worsening peak aerobic performance [227].

GENITOURINARY SYSTEM

Aging changes in the genitourinary system increase the older person's risk of urinary incontinence, urinary tract infection, erectile dysfunction, and dyspareunia.

Bladder — The prevalence of urinary incontinence increases with age. Until age 80, incontinence is more common in women than men, but the prevalence differences by sex subside after age 80 [228]. In healthy men aged 70 to 90, peak urine flow rate was unchanged over a five-year period while postvoid residual (PVR) bladder volume increased about 25 percent [229]. Urinary incontinence is related to decreases in detrusor muscle contractility, maximum bladder capacity, maximum flow rate, and the ability to withhold voiding, and an increase in PVR [230]. In cross-sectional data, the detrusor power decreased 30 percent in

healthy older women; a smaller decrease was seen in men [231]. These functional changes are due in part to decreased innervation of the detrusor muscle [232] and in part due to changes in the brain [233]. The common triad of increased PVR, reduced ability to withhold voiding, and nocturia have been attributed to decreased central nervous system dopamine rather than a local bladder-centric issue [234]. (See "[Urinary incontinence in males](#)" and "[Female urinary incontinence: Evaluation](#)".)

Beta-3 adrenergic agonists are used to reduce bladder contractions. When bladder smooth muscle from older adults is compared with middle-aged adults, there was no alteration in number of beta-3 receptors, but the response to CL316,243 was attenuated in the old muscle [235]. Whether this finding is applicable to the approved beta-3 agonists like [mirabegron](#) is unclear.

Withdrawal of estrogen in women results in decline in urethral length as well as decreased maximal urethral closure pressure. The urethra becomes a less effective barrier from bacterial contamination with age, especially in women [236]. Topical estrogens, especially in addition to pelvic floor exercises, may lead to restoration of urethral function [237].

It is increasingly clear that the urinary bladder supports as rich a microbiome as the gut. The urine is not truly sterile and, in women, the species of this microbiome are altered with menopause [238]. Whether the bladder microbiome has causative roles in age-related changes in bladder function is unclear.

Male reproductive system — Surveys of sexual activity among older men have found varying rates of activity. In one multinational survey, over 80 percent of men aged 60 to 69 and 70 percent of those aged 70 to 79 reported that they were sexually active [239]. By contrast, two representative samples from the United States found that 39 percent of men aged 75 to 85 years reported sexual activity [240]. The older penis needs greater stimulation to attain an erection, spontaneous erections are less frequent, erections are less firm, refractory times between erections (or ejaculations) become prolonged, and ejaculation is less forceful with smaller ejaculate volumes. These changes are the sum of age-related neurologic, vascular, and endocrinologic changes [241]. (See "[Epidemiology and etiologies of male sexual dysfunction](#)".)

A gradual decline in male reproductive ability occurs with age. Germ cells are formed continually, but sperm production decreases. The sperm from older testes have an increased frequency of chromosomal abnormalities and impaired motility and decreased ability to fertilize even when administered by intrauterine artificial insemination [242]. The seminiferous tubules also degenerate and Leydig cell number decreases. Changes in the epididymis and seminal vesicles are characterized by deposition of pigmented granules in the epithelial walls and amyloid in the seminal vesicle wall, which may be associated with amyloid deposition elsewhere in the body [243].

Enlargement of the prostate gland occurs with age. Prostate gland hyperplasia is discussed in detail separately. (See ["Epidemiology and pathophysiology of benign prostatic hyperplasia"](#) and ["Clinical manifestations and diagnostic evaluation of benign prostatic hyperplasia"](#).)

Female reproductive system — The ovary ages with a decline in oocyte numbers as women enter their late fourth decade, and menopause (amenorrhea for 12 months after the final menstrual period) ensues at an average age of 51 years. In women of advanced age undergoing in vitro fertilization (IVF) treatment (defined here as after 45), the implantation, clinical pregnancy, and live birth rates all occur at reduced frequency [244]. Changes related to menopause and estrogen depletion are discussed separately. (See ["Menopause: Clinical features and diagnosis"](#).)

Once menopause has occurred, more gradual age-related postmenopausal processes are seen. The vagina loses elasticity. The clitoris, like the older penis, needs greater stimulation and becomes less engorged [245]. Subcutaneous fat in the pelvis is lost. Vaginal dryness and atrophy are mostly estrogen-dependent but may be compounded by age-related diminished blood flow to the vagina. Cervicovaginal secretions, especially during arousal, become sparser. Vaginal pH rises, allowing colonization by enteric microflora and a loss of lactobacillus bacteria among other changes in the microbiome correlating with a decrease in amino acid availability in the post-menopausal vagina. The vaginal environment becomes much more proinflammatory using both cellular and cytokine markers [246]. (See ["Overview of sexual dysfunction in females: Epidemiology, risk factors, and evaluation"](#).)

MUSCULOSKELETAL SYSTEM

Muscle — Although there is great variability, muscle mass decreases in relation to body weight by approximately 30 to 50 percent in adults. The loss is not linear, but it accelerates with increasing age. Sarcopenia, age-related loss of muscle mass and strength, is defined as a decrease in appendicular muscle mass two standard deviations below the mean for young healthy adults [247]. Sarcopenia is an independent risk factor for mortality in longitudinal studies [248] and is found in as many as 50 percent of those over 80, depending on the population assessed and the definition of sarcopenia applied [249].

In addition to muscle mass loss, muscle quality decreases with infiltration of fat and connective tissue into the old muscle [250]. The presence of intramuscular and intermuscular fat has been termed "myosteatorsis." Myosteatorsis at the thigh has been associated with decreased strength, slower gait speed, and decreased survival in the AGES-Reykjavik study [251,252]; a mortality relationship was not seen with the calf myosteatorsis [253]. Critically, myosteatorsis needs to be taken in context, as exercise-trained individuals acquire muscle fat as a ready energy supply [254], and in that setting increased muscle fat is associated with better performance and exercise training in older people [255].

The loss of muscle mass is not uniform; in general, the loss from the legs is greater than from the arms. Type I slow-twitch fibers are less affected by age than fast-twitch fibers. The loss of fast twitch muscle with age can be prevented by lifelong strength training, but not endurance training [256]. In any muscle bundle, the size of the myofibrils decreases, followed by the number of myofibrils. Innervation of skeletal muscle decreases in men over 50; the number of motor units in any given muscle decreases with a compensatory increase in motor unit size. While this synaptic remodeling occurs at all ages, the "new" neuromuscular innervations are unstable [257]. An inability to finely modulate the stimulation rate may contribute to the loss of strength [258]. Some have implicated motor neuron changes as the primary cause of sarcopenia [259]. The loss of muscle contributes to age-related insulin resistance, age-related changes in body composition, and volumes of distribution for water-soluble drugs.

The presence of atrophied or partially or completely denervated muscle fibers can be seen on cross-section examination of muscle from an older person. Time to peak tension with ankle dorsiflexion is slowed, as is time to muscle relaxation [260]. Strength also decreases dramatically with age, partially explained by loss in muscle mass. From age 30 to age 80, a typical person's grip strength decreases 60 percent; however, activity plays an important mitigating role. Overall, lower-extremity strength is lost at a faster rate than upper-extremity strength; activity may decrease the rate of decline but will not completely prevent it [250]. The net result is that strength loss is greater than muscle mass loss, with strength loss being a better predictor of disability and mortality [261]. The older muscle is more easily fatigued as well [262].

The recovery of older muscle after injury is slowed, and frequently incomplete [263], perhaps related to a defect in satellite cells' ability to repopulate muscle [264]. This satellite cell defect is equally apparent in trained and sedentary older people [265]. Some of this impairment in recovery is locally mediated, perhaps by nerve regeneration, as old muscles transplanted into young animals regenerate fairly well, while muscles taken from young animals and transplanted into old ones do not regain mass or generate force as effectively [266]. Additionally, myostatin may be increased, but no age effect on circulating myostatin levels were seen with a highly specific assay in healthy men aged up to 85 [267]. Myostatin, a member of the myokine family, has a potent effect to decrease muscle protein synthesis and neutralizing antibodies to myostatin have been administered to people over 75 who were weak and had recent falls [268]. Although the antibody increased lean body mass, there was no effect on functional measures at 24 weeks.

Skeletal muscle from older adults shows altered energetics [269]. The decrease in enzyme activity of glycolytic enzymes is greater than that of oxidative enzymes. Physical activity plays a significant role in the decrease in these enzyme activities, and older adults with high physical activity have isolated mitochondrial function that is not significantly different from 25-year-olds [270]. Similarly, the in situ, noninvasive assessment of recovery of

phosphocreatine was reduced in typical older adults and was not different from the young in the physically active older adults [270]. In older animals, an acute bout of exercise is associated with relative hypoperfusion of the most oxidatively active exercising muscles [271].

Age-related hormonal changes in growth hormone, androgen, and possibly others may contribute to the age-associated alterations muscle mass and function. Parabiosis studies (where old and young syngenic mice are connected so that they share circulations) suggest circulating factors may restore many of these age-associated decrements in muscle [272,273]. Additionally, proinflammatory cytokines increase with age and stimulate the rate of skeletal muscle protein degradation [250]. The effects of agents that stimulate muscle growth, especially insulin-like growth factor 1 (IGF-1), may be mediated by neuromuscular effects, suggesting that both hormonal and neuronal approaches to preventing sarcopenia may be efficacious [259].

Bone — Aging increases the probability of fracture and the rate of repair is slowed, once fracture occurs. The increased proinflammatory environment in healthy older adults promotes bone loss. Anatomically, the weightbearing cortical bones lose substance from the endosteal surface. Computed tomography (CT) or MRI indicate that the marrow lumen of the femur is larger, the cortex thins, and fat fills much of the marrow cavities. The aging loss of mineral occurs in both cortical (peripheral skeleton) and trabecular (axial skeleton) bone. Trabecular number decreases and the distance between trabeculae increases with age in healthy aging women [274]. There is a progressive decline in osteoblast number and activity, but osteoclasts remain unchanged with age. Precursor cells for osteoblasts remain constant in number after age 30, but their function declines [275], with an enhanced tendency to become adipocytes rather than bone forming [276]. Whether or not the fundamental defect is in the bone marrow microenvironment or the precursor cells themselves remains uncertain [277].

Overall, the decline in bone mass is approximately 0.5 percent per year in healthy older people [278]. Age-related changes in women are compounded by menopausal changes in bone mass and function. Vitamin D deficiency including the impaired 1-alpha-hydroxylation of vitamin D by the aging kidney, common in older people, further accelerates bone loss. Perhaps because of shared mechanisms, osteoporosis and sarcopenia occur together with high frequency [279]. Their cooccurrence, osteosarcopenia, is increasingly recognized in older adult populations. Weightbearing exercise (walking) appears to have beneficial impact on both, but walking plus resistance exercise had a greater benefit [280]. (See "[Pathogenesis of osteoporosis](#)".)

Weightbearing exercise is frequently reduced in older adults, contributing to a negative calcium balance and loss of bone mineral [281]. Increasing weightbearing time or increasing loading forces may increase bone mineral and prevent age-related bone loss [282].

Bones from older adults have microanatomical changes that make them weaker than the amount of mineralization would predict. These changes include widening of the Haversian canals and thinning of the osteon wall [283]. Once bones fracture, the repair mechanisms are also impaired in aging. In older animals, fractures produce less local blood vessel formation and less osteogenic differentiation of progenitor cells and require at least twice as long to regain prefracture biomechanical properties, including strength, than in younger adult animals [284]. Cells isolated from old bones are less responsive to vitamin D than young ones. The matrix in old individuals may stimulate less bone formation than that of younger people. This suggests that growth factors (eg, IGF-1) may be deficient or inhibitory factors may be present in the old matrix. Supplementation with vascular endothelial growth factor (VEGF), parathyroid hormone, vitamin D and calcium, statins, and some of the bone morphogenic proteins have all shown promise in facilitating bone healing in various experimental paradigms [285]. This is true not only for clinically apparent fractures but also for the micro-cracks (microscopic disruptions in mineral or organic matrix), which increase dramatically with age. This micro-damage stimulates the repair mechanisms and further accentuates the age-related imbalance between bone deposition and resorption [286]. The microdefects may also make bone weaker than would be expected from just the bone density [287].

CENTRAL NERVOUS SYSTEM

Anatomical and physiologic changes — The volume of the brain decreases approximately 7 cm³ per year after age 65, with greatest loss in the frontal and temporal lobes [288] and greater loss of white matter than grey matter in cognitively normal older adults [289]. Cerebral blood flow decreases heterogeneously by 5 to 20 percent, with deterioration of mechanisms that maintain cerebral blood flow with fluctuation in blood pressure [290]. In addition to older age, males have higher cerebrovascular impedance measured noninvasively, which may promote brain hypoperfusion [291].

Age-related neuronal loss is most prominent in the largest neurons in the cerebellum and cerebral cortex. The hypothalamus, the pons [292], and the medulla [293,294] have modest if any neuron or volume losses with normal aging. Age-related neuron dropout is likely due to apoptosis (ie, programmed cell death) rather than inflammation, ischemia, or another mechanism [295]. Age also affects neurons that persist, with loss of the dendritic tree, shrinkage of processes, and decrease of synapses [296]. Such changes may contribute more to the age-related loss of brain volume than the loss of neurons. In some areas, however, the dendritic connections may increase, perhaps as a result of repatterning of the brain invoked to compensate for cellular dropout. Neurons continue to form new synapses, and new neurons are formed throughout the lifespan, but the rates of loss are greater than the gains [297].

Lipofuscin accumulates in certain areas of the brain, particularly the hippocampus and frontal cortex, but the impact of lipofuscin on function is unknown [298]. Neurofibrillary tangles and senile plaques occur in certain areas of the brain in normal aging but to a lesser extent than in Alzheimer disease. More than 50 percent of cognitively normal individuals over age 85 have sufficient plaques/tangle burden to make a pathologic diagnosis of Alzheimer disease [299]. Thus, interpretation of beta amyloid seen on amyloid imaging in individuals of advanced age poses a challenge [300]. Similar issues may be raised for tau imaging [301]. For both beta amyloid and tau, there were thought to be brain regions where their accumulation started but, with more recent studies, there is evidence of multiple brain regions with tau and beta amyloid signals in cognitively unimpaired older adults, correlating with age more than cognition [302].

Multiple nonhomogenous changes in brain enzymes, receptors, and neurotransmitters occur with age. Acetylcholine availability decreases due to decrease in cholinergic and muscarinic neurons, and reduced release and synthesis of acetylcholine [296,303]. As well, dopamine and corresponding receptors in the striatum and substantia nigra may be decreased in normal aging [304]. Interestingly, dopamine may facilitate episodic memory persistence [305], and providing dopamine as L-DOPA to normal old brains can improve performance on some cognitive tasks [306].

Age-related changes seen on functional brain scanning with fluorodeoxyglucose positron emission tomography (FDG-PET) are heterogeneous [307]. For many tasks, the old brain seems to work harder than the young one, recruiting more neurons and with higher energetic expense as shown by PET scan [308]. For a simple recall test of letters, a greater volume of activation was seen in old brains. The amount of brain activated for a given task, including bilateral activation, may be an insight into cognitive reserve, brain vulnerability, and tradeoffs in efficiency made to maintain cognitive function with age.

Brain connectivity, as assessed with functional MRI, is altered by normal aging. The default mode network is modified so that the connections are less robust but are still greater than the changes seen in Alzheimer disease [309]. In the absence of disease, the loss of default mode connectivity is associated with decreased memory performance [310]. In contrast to the decreased within-network connectivity, there is increased between-network connectivity which may contribute to decreased efficiency [311].

Cognitive and behavioral changes associated with normal aging — Certain memory performances on cognitive testing, like procedural, primary, and semantic memory, are well preserved with age [312]. Skills, ability, and knowledge that are overlearned, well-practiced, and familiar, like vocabulary or general knowledge, remain stable or improve up to 0.2 standard deviations per decade through the seventh decades, but even these processes can begin to decrease with further aging [313]. Older people may be more accurate in judging

distances than younger people [314]. The ability to recognize familiar objects and faces, as well as to maintain appropriate visual perception of objects, remains stable over the lifetime.

Episodic and working memory and executive function are the specific domains of cognition most affected by "normal" aging [315]. These are late-life changes, occurring after the sixth decade, and have a linear or accelerating decline with further aging [316]. Processing speed decreases with age and can have a global effect on the testing performance of other neurocognitive domains in any timed test [317]. This may contribute to the slower speech production of older adults [318].

Executive function, the higher-level organizational abilities such as planning, monitoring, inhibition, and control of action, is critical to engagement in purposeful, independent, and self-preserving behavior and is necessary for an older person to successfully manage their own medical illnesses. Executive function declines with age and more dramatically after age 70 [319].

Attention span decreases with even simple attentive tasks [320]. In particular, there is a decrease in the ability to focus on a task in a busy environment and the ability to perform multiple tasks at one time [319]. For example, asking an older person to say the alphabet or count backwards slows their gait speed by 10 to 20 percent and may increase their fall risk [321]. These impairments, similar to processing speed, may lead to decreased testing performance in other neurocognitive domains.

Problem-solving, reasoning about unfamiliar things, processing and learning new information, and attending to and manipulating one's environment show a steady decline (by approximately -0.02 standard deviations per year) after peaking around age 30. Language abilities (verbal fluency and the ability to name objects) demonstrate some late-life decline, particularly after age 70.

The magnitude of aging is tempered by the effects of social deprivation, environmental changes, smoking, alcohol, education, and other aspects of the exposome [322]. One of the less recognized is the effect of sleep on long-term cognition [323]. An article studied a Scottish population in Lothian, retesting them at age 70 on tests that they took as 11- or 12-year-olds. The comparison of their young age scores with those more than 50 years later are shown in the brush plot [324]. A sizeable minority scored better than they did when they were younger, but the heterogeneity is great.

Despite measurable changes seen on cognitive testing with normal cognitive aging, the successfully aging 95-year-old individual remains able to function in society, the workplace, and/or at home. Few real-life situations require performance at maximum levels, especially with time pressure or acquired knowledge. The impact of cognitive loss can often be compensated by noncognitive factors that do not decline with age [313].

Cognitive retraining — Novel neural challenges increase recruitment of additional brain regions in young healthy subjects; with repetition of the challenge, recruitment of these brain regions subsides and activity is seen in skill-specific regions [325-327]. Neural recruitment in the aging brain is used to accomplish less novel tasks. This process, referred to as "compensatory scaffolding," may be a strategy the brain uses to maintain function and cognition [328].

Repeated use of "compensatory scaffolding" by engaging in social, leisure, and cognitive activities (learning a new language, pursuing higher education) may decrease the risk of Alzheimer disease or delay its onset [329] and slow the progress of normal aging changes [330]. Specifically designed cognitive training activities for older adults have also been shown to decrease decline in ability to perform instrumental activities of daily living, per subject self-reports compared with controls [331]. In healthy volunteers, cognitive training can lead to gray matter volume increases in the "exercised" areas [332]. Whether these efforts improve compensation or prevent age-related decline in neurologic processes is uncertain.

SKIN

The normal aging of the skin leads to atrophy, decreased elasticity, and impaired metabolic and reparative responses. These changes are separate from those due to sun exposure, so-called "photoaging."

The epidermis becomes thinner, and the dermoepidermal junction flattens, resulting in increased fragility of the skin to shear stress [333]. Removing an adhesive dressing from an older person may dislodge the epidermis because the dermoepidermal junction is weaker than the bond between the skin and the dressing. Bleeding into the space between the dermis and epidermis occurs more frequently.

Loss of undulations at the dermoepidermal junction decreases the area available for nutrient transfer, including protective lipids in the stratum corneum. This results in dry skin (xerosis) and a compromise in the barrier function of the skin [334]. Epidermal turnover is slowed due to decreased division of keratinocytes and longer migration from the basal layer to the skin surface [335]. The epidermal cellular composition changes, with decreases in melanocytes and immunologically active Langerhans cells, and a 50 percent overall reduction in nail growth and reductions in sweat and sebaceous gland activity [336].

Additional changes associated with age rather than sun damage include the following:

- The dermis thins with decrease in vascularity and in the biosynthetic capacity of the resident fibroblasts [337]. These changes contribute to delayed wound healing. The amount of dermal collagen may be decreased by 75 percent with age [338], and the remaining collagen is fragmented and disarrayed.

- The elastic fiber network degenerates as elastin biosynthesis declines significantly after the fourth decade. Elastin fibers in aging abdominal skin are reduced in number, fiber diameter, volume fraction, and network connectivity [339]. Changes in the glycosaminoglycan macromolecules in the dermis lead to loss of hydration and decreased skin resilience [340].
- The ability to deliver heat to the skin for excretion is impaired, especially during exercise [341]. With the loss of rete pegs at the dermoepidermal junction and loss of dermal capillarity, the area for heat transfer to the epidermis is decreased. Loss of subdermal fat decreases insulation and the ability of older people to conserve heat. Tonic vasoconstriction in many older adults, as well as decreases in the amount of sweat produced by sweat glands and higher core temperature before sweat is produced, all contributing to impaired thermoregulation with age [342-344].
- Sensory perception of the skin decreases, particularly in the lower extremities [345]. Decreased sensation involves both touch, due to decreased Meissner's corpuscles [346], and low-frequency vibration, mediated by the Pacinian corpuscles [347].
- The skin plays a critical role in vitamin D synthesis. Ultraviolet rays convert 7-dehydrocholesterol to pre-vitamin D3 in the epidermis. Levels of 7-dehydrocholesterol decreased with age, thus decreasing the older person's capacity for vitamin D synthesis [348].
- Senescent cells accumulate in the skin of older people [349].
- The microbiome of the skin changes with age such that increased diversity is seen on the older skin [338] and age can be predicted from the microbiome with a mean absolute error of about five years [350].
- There is a decrease in subdermal fat. This loss of support contributes to the skin wrinkling and sagging as well as to increased susceptibility to trauma [351].

Topical administration of all-trans-retinoic acid (tretinoin) appears to reverse many of the age-related changes in sun-protected skin (inner thigh). After nine months of daily treatment with [topical tretinoin](#) cream 0.025 percent, the epidermis thickens, rete ridges become deeper, capillaries reappear, matrix proteins are redeposited, and collagen and elastin increased [352]. Thus, these age-related changes appear to be mutable.

The skin has its own microbiome with diversity in part dictated by the immune system, moisture, and sebum production. In older women, the diversity was greater than those 21 to 37 for the cheek and forehead [353]. The dominant organisms change as well, with older individuals having less *Propionibacterium*, for example.

Photoaging is the result of chronic sun exposure and recurrent damage by the sun's ultraviolet light. Photoaging, not physiologic aging, produces most of the cosmetically

undesirable changes in skin. Cellular dysplasia, atypical cells, a loss of polarity of the keratinocytes, and a significant disorganization in the epidermis are the result of photoaging. In the dermis, photoaging leads to elastosis, aggregates of amorphous elastic fibers, a decrease in collagen content, an increase in glycosaminoglycans, and a modest inflammatory infiltrate localized to the perivascular areas. The photoaged skin looks wrinkled, lax, yellowed, rough, and sometimes leathery. Photoaged skin has a higher tendency toward telangiectasias, and it is spottily hyperpigmented and hypopigmented. Photoaging changes are also partially reversible by topical treatment with retinoic acid.

SENSORY SYSTEM

Eye — The structure of the eye changes with age. Periorbital tissues atrophy; eyelids become more relaxed. The lower lid flaccidity may lead to ectropion (eyelid turns outward) or entropion (eyelid turns inward). Lacrimal gland function, tear production, and goblet cell function all decrease [354]. Even though tear production decreases, watering eyes becomes more common because tissue atrophy leads to displacement of the lacrimal punctum and less effective drainage.

The conjunctiva atrophies and yellows. The sensitivity of the cornea to touch declines by 50 percent. Deposition of cholesterol esters, cholesterol, and neutral fat in the cornea causes arcus senilis, an annular yellow-white deposit on the peripheral cornea. The presence of arcus senilis correlated with shorter lifespans in women in the Copenhagen City Heart Study [355]. The iris becomes more rigid, yielding a smaller, more sluggishly responsive pupil. The lens yellows, in part because of photo-oxidation in lens protein and an accumulation of insoluble protein. The yellowing of the lens causes decreased transmission of blue light [356].

Production of aqueous humor decreases and the vitreous humor and body also shrink. Separation between the liquid and solid components of the vitreous may be due to collagen changes and manifest as flashes of light [357]. The retina becomes thinner because of a loss of neurons. The retina may also become more inflammatory with increasing infiltration of macrophages and activation of glia cells [358]. Retinal blood flow also decreases with age in healthy adults over 65 by 20 percent compared with younger adults [359]. The relationship between the blood flow and the loss of neurons is unclear.

The changes in lens and iris lead to "presbyopia." The distance needed to focus near objects increases because of decreased lens elasticity and, to a lesser extent, weakening and loss of an effective angle of the ciliary muscle [360]. Presbyopia has gradual onset in the fourth decade with steady deterioration in static acuity (object at rest) and a more pronounced loss of dynamic visual acuity (ie, objects in motion). (See "[Visual impairment in adults: Refractive disorders and presbyopia](#)".)

The older eye adapts more slowly to changes in lighting conditions; the pupil becomes rigid and the lens more opaque. The rate of synthesis of photopigment slows with age, adding to slowed adaptation to lower light conditions [361]. The absorption of photons by photoreceptors is also decreased with age [362]. The retina becomes thinner, especially in the retinal nerve fiber layer. By contrast, the retinal pigment epithelium (RPE) thickens with age [363]. Lens alterations increase light scattering, making the older person more sensitive to glare. After lens removal, the glare threshold becomes normal.

Finally, contrast sensitivity declines, so older persons need increased color contrast to discriminate between target and background. This factor should be taken into account in the design of living environments.

Hearing — Age-related changes in the auditory system produce decrements in high-frequency hearing acuity and impaired speech recognition in noisy environments. The loss of hearing acuity may result in social isolation and increases risk for delirium during hospitalization, but there is uncertainty about the risk of dementia [364,365]. The cumulative effects of environmental or occupational noise confound interpretation of age effects. (See ["Etiology of hearing loss in adults"](#).)

With age, the walls of the external auditory canal thin. The cerumen becomes drier and more tenacious, increasing the risk of cerumen impaction in older people. Although the ossicular joints degenerate with age, sound transmission by the ossicles is well preserved.

The inner ear experiences at least five distinct changes that occur to varying degrees:

- Hair cells in the organ of Corti are lost, initially affecting those in the basal end of the cochlea that respond to the highest frequencies
- Neurons innervating the cochlear and in the auditory centers of the brain are lost
- The basilar membrane underlying the sensory apparatus stiffens and may calcify
- The capillaries of the stria vascularis (the source of endolymph) thicken
- The spiral ligament degenerates

Which of these five changes is dominant will define subgroups of age-related hearing. The net results are loss of hearing acuity, especially at higher frequencies (presbycusis); difficulty with speech discrimination; and problems localizing the sources of sound [366,367].

Some older individuals who say they cannot hear in fact are having difficulty understanding what is said. Many of the consonant sounds are in the higher frequencies (t, k, ch), and patients may not comprehend speech if they cannot appreciate those sounds. Strategically and practically, it may be better to rephrase a question that is not understood by an older person rather than repeat it in a louder voice, especially because pitch (frequency) is often raised when volume is increased. Additionally, older people may have difficulty discriminating target sound from background noise, adding to challenges of communicating in social

situations or noisy emergency departments [368]. In these patients, amplification of sound alone is not effective because both target and background are amplified. Finally, as hearing relies so much on cognitive processing, older adults, when they have their hearing restored (by cochlear implant), have initial gains but are less likely to have the subsequent gains in speech recognition that younger adults experience. This is attributed to a lack of neural plasticity [369]. Despite that, a small prospective study found that after cochlear implant, 38 percent of older adults with mild cognitive impairment improved and no longer tested as cognitively impaired [370].

Taste and smell — There are visible changes in the taste buds with age, though they have modest impact on the sense. Although the number of papillae on the tongue decreases with aging, neurophysiologic responses of individual papillae are minimally altered, and there is no relation between gustatory acuity and number of taste buds.

Loss of taste in older patients is in large part due to decreased olfaction rather than taste itself [40]. However, taste sensitivity also decreases with age, as shown by a study in which older patients required 30 percent higher concentrations of aspartame to detect this artificial sweetener [371,372].

Similarly, more salt (two- to threefold) needs to be added to tomato soup before it can be appreciated by an older person [373]. The effects of age on the tongue need not be uniform, with regions of deficient gustatory sense becoming more common with age.

The acuity of olfaction declines significantly with age. Detection thresholds are increased by more than 50 percent in healthy people by age 80; recognition of familiar smells decreases similarly, including the recognition of spoiled food and of a gas stove left on. The decrease in olfactory acuity after exposure to an odorant (habituation) is prolonged and of greater extent in older adults [374]. The cause of the decreased olfactory sense is unclear, but the sensation area decreases, the number of sensing neurons decreases, and the ability of the older person to replenish dying olfactory receptor neurons is compromised [373].

Decreased taste and smell sensation may result in decreased enjoyment of food and an age-related difficulty in sorting tastes of mixed or combined foods. The role of olfaction in maintaining appetite is critical; for example, some people with anosmia forget to eat. This is especially important because the sensation of hunger after an overnight fast is roughly 25 percent lower in older adults than in young adults [375].

IMMUNE SYSTEM

Of all the changes that occur with age, decrements in immune functions are among the most critical, contributing to the increased frequency of infections, malignancies, and autoimmune

disorders. These changes are mentioned briefly here and reviewed in more detail separately. (See ["Immune function in older adults"](#).)

Immunosenescence, or the aging of the immune system, does not impact all immune processes equally. Some of the responses that are most affected by age include the ability of lymphocytes (both B and T cells) to work in concert to generate effective immune responses upon exposure to new antigens, in the form of either infections or vaccinations.

An important concept in immunosenescence is that of loss of precise regulation of inflammatory processes. Older adults display cytokine profiles that are consistent with a chronic, low-level inflammatory state, which is sometimes referred to as "inflammaging" [376]. This chronic low-grade inflammatory state contributes to age-associated morbidity and mortality and many of the age-related changes described above. (See ["Immune function in older adults"](#).)

MOLECULAR BASIS OF AGING

The molecular basis for age-related physiologic changes is a subject of intense active investigation. The process of natural selection (in which genes that promote beneficial conditions in early or reproductive years are selected for, and genes associated with harmful conditions are selected against), does not play a significant role in later life. Despite the lack of selection for late-life genes, some processes that may provide benefit early in life could be detrimental later in life, so-called antagonist pleiotropy.

For more than 50 years, the most robust means to increase survival and modify many age-related changes was to use caloric restriction. Reducing caloric intake by 20 to 40 percent in some of the most common strains of laboratory mice and rats increased median and maximum lifespan by 20 to 50 percent but did not increase survival in approximately 50 percent of tested mouse strains [377]. Studies with nonhuman primates have been less promising, and, despite having lower cholesterol, better insulin sensitivity, etc, survival has not been increased in one primate study and not another, leading to the conclusion that dietary composition is an additional critical factor [378-380].

Several findings suggest that shortening of the telomere (nucleoprotein end caps on chromosomes) is involved in increased vulnerability of aging cells to DNA damage and dysregulation [381-383]. The shortened telomeres, as well as other replicative dysregulation, may lead to inadequate replacement of damaged or dead cells from their respective precursor cell populations. Interestingly, many of these resting precursors cells start to differentiate along adipocyte-like pathways, rather than into other tissue types [18].

Subpopulations of adipocytes, hepatocytes, fibroblasts, and other cells may enter senescence with aging and develop the senescence associated secretory phenotype (SASP) [384]. SASP

cells have the potential to release proinflammatory cytokines and other potentially harmful factors as well as modify the activity of local normal cells [385]. The SASP may contribute to "inflammaging" [386]. Agents that kill the senescent cells (senolytics) or attenuate their adverse effects ("senomorphics") improve various aspects of aging in animal models and early studies in humans [387,388]. T cells have been engineered to target and destroy senescent cells [389]. Other hypotheses implicate the p53 gene that is activated when DNA is damaged. This gene activation may mediate several molecular processes affecting cell function and viability: normal cell growth and division is halted, with apoptosis for cells that rapidly turnover; peroxisome proliferator-activated receptor gamma coactivators PGC_{1-alpha} and PGC_{1-beta} are repressed and lead to loss of muscle mitochondria and buildup of free radicals with loss of antioxidant defenses [390]. Importantly, replicative senescence is not the only nor even dominant path to senescence; chromosomal damage and certain signaling pathways lead to senescent cells that may have relatively long telomeres.

The mammalian target of rapamycin (mTOR) pathway regulates nutrient distribution and is believed to play an important role in the ability of caloric restriction to extend lifespan. Rapamycin has been shown to produce longevity in mice [391].

The processes common to aging in people as well as animal models have been called the pillars or hallmarks of aging [392]. Lopez-Otin expanded the hallmarks to 12: genomic instability, telomere attrition, epigenetic alterations, loss of proteostasis, disabled macroautophagy, deregulated nutrient sensing, mitochondrial dysfunction, cellular senescence, stem cell exhaustion, altered intercellular communication, chronic inflammation, and dysbiosis. This has allowed researchers to organize their approaches with the expectation that modifying one pillar will have a multitude of effects across the other pillars. As an example, most cardiovascular diseases of the older adult have strong association with abnormalities or dysfunction for every one of the pillars of aging [393].

LONGEVITY

Longevity predictions are important in many aspects of medical care. Limited survival will impact the benefits of initiating medications or performing procedures or screening tests. Decision-making regarding the appropriateness of a specific intervention (eg, "Should this older person be given a bisphosphonate to prevent osteoporosis?") requires recognition of both the likely survival time for an individual and how long it may take for the intervention to have effect.

One classic study provides survival data on individuals in the United States, stratified into quartiles by survival for each five-year period above age 65 years [394]. The absence of significant comorbid conditions, or the presence of superior functional status for age, identifies older adults who are likely to live longer than average. Conversely, individuals with

functional dependencies in activities of daily living and/or significant comorbidity (eg, heart failure, end-stage kidney disease, oxygen-dependent chronic obstructive lung disease) have a life expectancy substantially below the average for their age.

Age-related modification of risk-benefit relationships has broad applicability. In addition to the documentation of the broad range of life expectancy for a given age, a table has been developed to measure the likely benefit against the potential risk ([table 2](#)).

For example, a typical 85-year-old man who has a symptomatic inguinal hernia is likely to have the problem for another 4.7 years and thus may warrant a discussion of surgical repair. Note that the tables are based on a United States White population and may not be generalizable to other groups. Also, the tables do not account quality of life for the specified years. Both of these issues may critically modify the interpretation of the tabular data. These concepts are integrated into tools at "eprognosis," which makes the data more specific to the individual [\[395\]](#).

Successful aging — The term "successful aging" is used to identify older individuals who are free from chronic disease and continue to function well into old age, both physically and cognitively. The term "essentially healthy" identifies those with no acute disease, no recent history of cancer, and well-controlled chronic disease. "Exceptionally healthy" identifies older adults who take no medications, have no chronic disease, are normotensive, and have a normal body weight. Psychosocial and genetic factors contribute to successful aging as well as longevity.

Predictors of high functional status in both physical and cognitive domains were evaluated in the longitudinal Cardiovascular Health Study, following 1677 participants for 14 years (median age 85, range 77 to 102, at study endpoint) [\[396\]](#). Although all participants showed functional decline over time, 53 percent remained functionally intact, and this group had a higher baseline health profile and lower vascular disease risk. Greater physical impairment was found in women and in those with greater weight. Cardiovascular disease and hypertension were predictors for both cognitive and physical impairment.

In another longitudinal study of nearly 6000 British civil service employees followed for 17 years, successful aging was identified in 12.8 percent of men and 14.6 percent of women at follow-up [\[397\]](#). The strongest predictor of successful aging was socioeconomic status at

Life expectancy by age

	Ages in years					
	70	75	80	85	90	95
Women - Years of expected life						
Top 25 th percentile	21.3	17	13	9.6	6.8	4.8
50 th percentile	15.7	11.9	8.6	5.9	3.9	2.7
Lowest 25 th percentile	9.5	6.8	4.6	2.9	1.8	1.1
Men - Years of expected life						
Top 25 th percentile	18	14.2	10.8	7.9	5.8	4.3
50 th percentile	12.4	9.3	6.7	4.7	3.2	2.3
Lowest 25 th percentile	6.7	4.9	3.3	2.2	1.5	1

[Life expectancy by age](#)

Table 2 - larger image below

midlife. After adjustment for socioeconomic factors, not smoking, diet, exercise, moderate alcohol intake (in women), and work support (in men) predicted healthy aging.

The effect of genetics on longevity and life expectancy has been explored in observational studies and in animal experiments. As noted above, twin studies suggest that 25 percent of longevity is genetic [3]. However, the importance of genetics is substantially larger at the extremes of longevity and may be greater for men than for women. In the New England Centenarian Study, male siblings of centenarians had a 17 times greater chance of living to age 100 compared with birth cohorts, compared with eight times greater for females [398].

A polymorphism in the Cholesterol Ester Transfer Protein gene has been identified that is associated with successful aging, increased longevity, preserved cognition, less cardiovascular disease, and larger-sized high-density lipoprotein (HDL) particles in Ashkenazi Jews [399]. Animal models suggest the potential for gene manipulation to prolong longevity. For example, the median and maximum survival of the nematode *Caenorhabditis elegans* can be increased threefold by mutating the *daf-16* gene, a key regulator of insulin/insulin-like growth factor (IGF)-like pathways in the worm [400].

Environmental factors are likely to interact with genetics. In Italy, the ratio of female to male centenarians is quite variable, ranging from 2:1 in Sardinia to 7:1 in Northern Italy, suggesting a gene-environment interaction [401]. Okinawa, Japan has the highest concentration of centenarians in the world. The longevity there has been attributed to a "caloric restriction with optimal nutrition" diet, similar to diets that increase longevity by 50 percent in mice and rats [402]. A genetic component is also possible, based on family studies of Okinawan centenarians [403]. Extreme female longevity may be less dependent on genetics than male longevity and more related to a healthier lifestyle and more favorable environmental conditions [401].

The heterogeneity of older people is remarkable and important to consider when reading this chapter. The rates of changes and compensations for the multitude of processes altered by aging are very variable, and, similarly, the extents and interactions of the pillars of aging are also variable. The classic saying "when you have seen one 85-year-old, you have seen one 85-year-old" encapsulates that variability.

Finally, one reason for the heterogeneity described above is the differences in aging between males and females. This has been underemphasized in this chapter for brevity, but, at least for cardiovascular aging, there are important and fundamental differences between aging in males and in females [404]. The differences lead to heart failure with preserved ejection fraction being a much greater risk for females. Similarly, for agents given to mice to increase their longevity, many of them have had sex specificity, increasing survival of males, not females, and vice versa [405,406]. The mechanisms and implications of these differences need to be addressed.

INFORMATION FOR PATIENTS

UpToDate offers two types of patient education materials, "The Basics" and "Beyond the Basics." The Basics patient education pieces are written in plain language, at the 5th to 6th grade reading level, and they answer the four or five key questions a patient might have about a given condition. These articles are best for patients who want a general overview and who prefer short, easy-to-read materials. Beyond the Basics patient education pieces are longer, more sophisticated, and more detailed. These articles are written at the 10th to 12th grade reading level and are best for patients who want in-depth information and are comfortable with some medical jargon.

Here are the patient education articles that are relevant to this topic. We encourage you to print or e-mail these topics to your patients. (You can also locate patient education articles on a variety of subjects by searching on "patient info" and the keyword(s) of interest.)

- Basics topic (see ["Patient education: Sex as you get older \(The Basics\)"](#))

SUMMARY AND RECOMMENDATIONS

- **Physiologic changes** – Aging, characterized by progressive changes associated with increased susceptibility to many diseases, is influenced by genetic factors, lifestyle choices, and environmental exposures. Several overarching physiologic principles characterize aging: loss of complexity as seen in less variability in heart rate responses, altered circadian patterns, and loss of physiologic reserves needed to cope with challenges to homeostasis. (See ["Age-associated physiologic changes"](#) above.)
- **Hematopoietic system** – Functional bone marrow reserves are reduced, with delayed response to blood loss or hypoxia. White blood cell function is impaired and myelotoxicity from chemotherapy is often increased. Advancing age leads to a procoagulant state, with increased platelet responsiveness and levels of clotting factors. (See ["Hematopoietic system"](#) above.)
- **Gastrointestinal system** – Reflux esophagitis, due to altered contractions and sphincter tone, is common with age and may affect nutrition. *Helicobacter pylori* infection is common, and sensitivity to gastric irritants such as nonsteroidal antiinflammatory drugs (NSAIDs) is increased. Colonic changes include motility changes resulting in constipation, decreased visceral response to bowel perforation or ischemia, colon diverticula, and increased risk for colon cancer. (See ["Gastrointestinal tract"](#) above.)
- **Renal system** – Kidney mass and function decline, with decrease in creatinine clearance and ability to maximally dilute urine or excrete an acid load. The older kidney is more

prone to nephrotoxicity related to medications or intravenous contrast. (See '[Renal system](#)' above.)

- **Cardiovascular system** – It is difficult to isolate the impact of age alone on the cardiovascular system, since age increases risk for hypertension, coronary artery disease, and a more sedentary lifestyle. There is an age-related decrease in maximum heart rate, as well as the compensatory response of left ventricular ejection fraction (LVEF) to exercise. The annulus of both the aortic and mitral valve thicken, with development of valvular calcific deposits. (See '[Cardiovascular system](#)' above.)
- **Respiratory system** – Approximately one-third of surface area per volume of lung tissue is lost over the lifespan, with increase in anatomic dead space increases and decrease in functional reserves. Age increases the alveolar-arterial (A-a) oxygen gradient, more marked in the supine than sitting position. Cough is less vigorous in the older person and mucociliary clearance is slower. (See '[Respiratory system](#)' above.)
- **Genitourinary system** – Aging changes in the genitourinary system increase the older person's risk of urinary incontinence, urinary tract infection, erectile dysfunction, and dyspareunia. (See '[Genitourinary system](#)' above.)
- **Musculoskeletal system** – With aging, muscle mass decreases in relation to body weight, leading to impaired motility and balance, as well as age-related increased insulin resistance and changes in the volume of distribution for water soluble drugs. Aging increases the probability of fracture and slows the rate of fracture repair. Increasing weightbearing may increase bone mineral and prevent age-related bone loss. (See '[Musculoskeletal system](#)' above.)
- **Skin** – The normal aging of the skin leads to atrophy, decreased elasticity, and impaired metabolic and reparative responses. Photoaging, not physiologic aging, produces 90 percent of the cosmetically undesirable changes in skin. (See '[Skin](#)' above.)
- **Sensory system** – Presbyopia is due to age-related changes in the lens and iris. Age-related changes in the auditory system produce decrements in high-frequency hearing acuity and impaired speech recognition in noisy environments. Loss of taste in older patients is in large part due to decreased olfaction rather than taste itself. Decreased taste and smell sensation may result in decreased enjoyment of food and result in nutritional deficiencies. (See '[Sensory system](#)' above.)
- **Immune function** – Decrements in immune functions with aging are among the most critical changes, contributing to the increased frequency of infections, and malignancies. Immunosenescence is associated with loss of regulation of inflammatory processes. Older adults display cytokine profiles that are consistent with a chronic, low-level

inflammatory state, sometimes referred to as "inflammaging." (See ["Immune function in older adults"](#).)

ACKNOWLEDGMENT

We wish to acknowledge the input of Nicolin Neal, MD, in the preparation of sections in this topic content.

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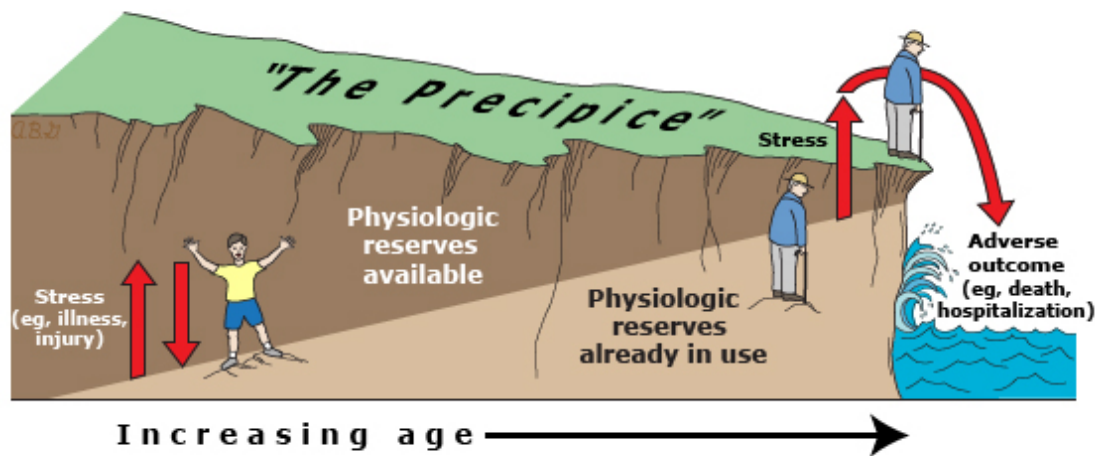
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GRAPHICS

Figure 1: Homeostenosis



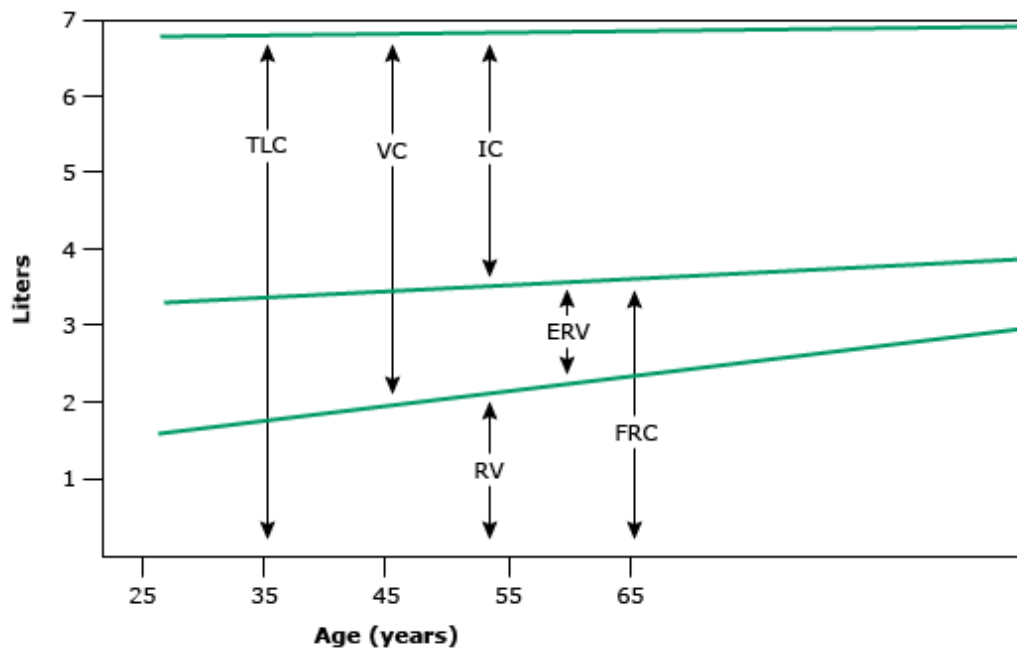
Based on information from: Taffet GE. Physiology of aging. In: Cassel CK, Leipzig RM, Cohen HJ, et al [eds]. *Geriatric Medicine: An Evidence-Based Approach*, 4th ed. New York, Springer, 2003.

Table 1: Renal changes with aging

Anatomic
Decrease in renal mass, mostly from cortex
Increased renal fat and fibrosis
Sclerosis of cortical nephrons with longest loops of Henle
Functional
Decreased renal blood flow
Impaired vasodilation in response to dopamine
Decreased creatinine clearance
Impaired sodium excretion and conservation
Decreased potassium excretion and conservation
Decreased concentrating and diluting capacity
Impaired excretion of acid loads
Decreased serum renin and aldosterone
Altered intrarenal nitric oxide actions
Increased dependence on renal prostaglandins to maintain intrarenal perfusion
Decreased vitamin D activation
Increased vulnerability to dye, ischemia, or other insults
Impaired recovery after insults

Data from: Taffet GE. Physiology of aging. In: Geriatric Medicine: An Evidence-Based Approach, Cassel CK, Leipzig RM, Cohen HJ, et al (Eds), 4th ed, Springer, New York 2003.

Figure 2: Pulmonary function with age



TLC: total lung capacity; VC: vital capacity; IC: inspiratory capacity; ERV: expiratory reserve volume; RV: residual volume; FRC: functional residual capacity.

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Table 2: Life expectancy by age

	Ages in years					
	70	75	80	85	90	95
Women - Years of expected life						
Top 25 th percentile	21.3	17	13	9.6	6.8	4.8
50 th percentile	15.7	11.9	8.6	5.9	3.9	2.7
Lowest 25 th percentile	9.5	6.8	4.6	2.9	1.8	1.1
Men - Years of expected life						
Top 25 th percentile	18	14.2	10.8	7.9	5.8	4.3
50 th percentile	12.4	9.3	6.7	4.7	3.2	2.3
Lowest 25 th percentile	6.7	4.9	3.3	2.2	1.5	1

Data from: Walter LC, Covinsky KE. Cancer screening in elderly patients: a framework for individualized decision making. *JAMA* 2001; 285:2750.

Contributor Disclosures

George E Taffet, MD Stock Options: Animatus Biosciences [Stem cells]. Grant/Research/Clinical Trial Support: Indus Instruments [Activity and cardiac function]. All of the relevant financial relationships listed have been mitigated. **Kenneth E Schmader, MD** No relevant financial relationship(s) with ineligible companies to disclose. **Jane Givens, MD, MSCE** No relevant financial relationship(s) with ineligible companies to disclose.

Contributor disclosures are reviewed for conflicts of interest by the editorial group. When found, these are addressed by vetting through a multi-level review process, and through requirements for references to be provided to support the content. Appropriately referenced content is required of all authors and must conform to UpToDate standards of evidence.

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